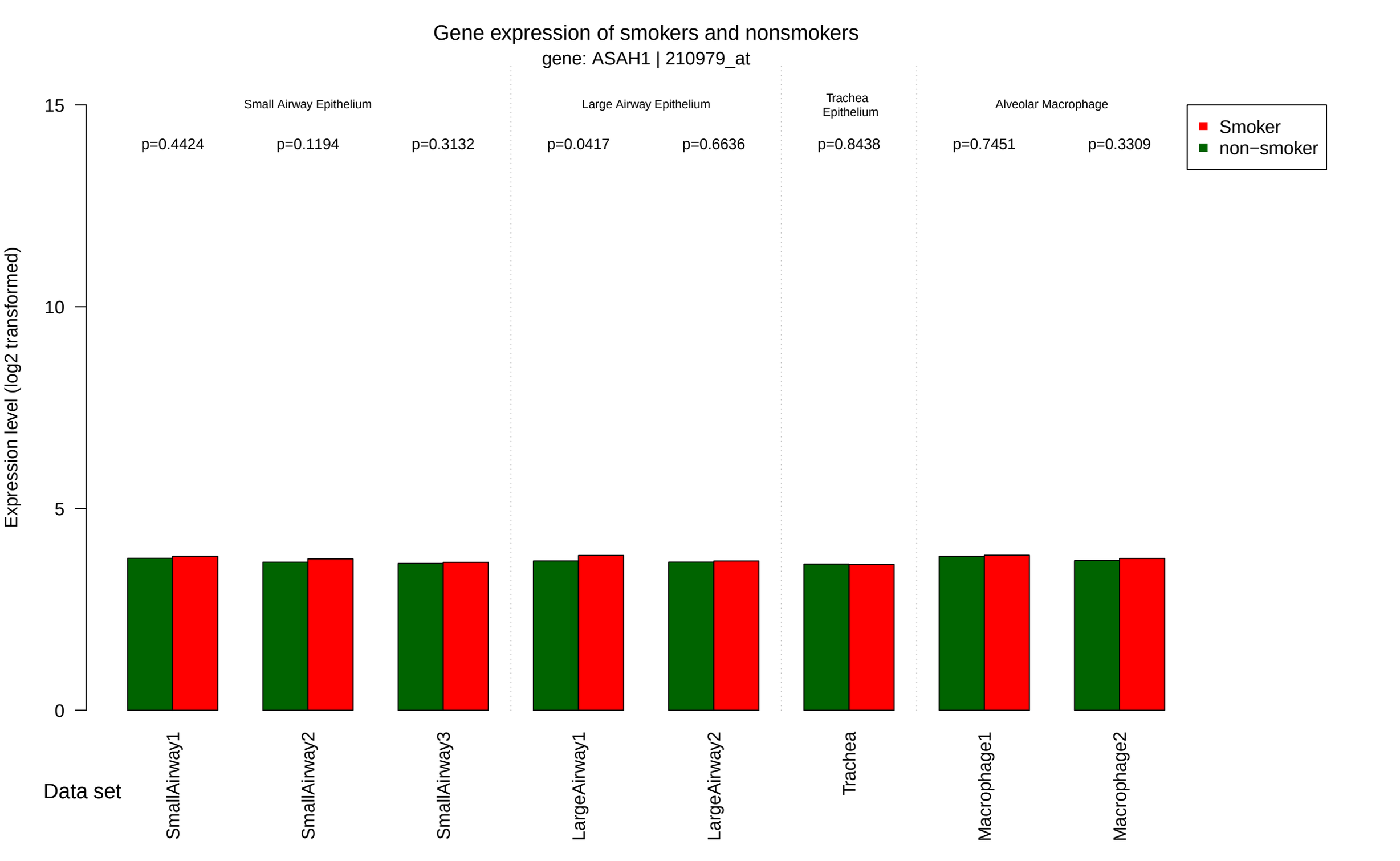
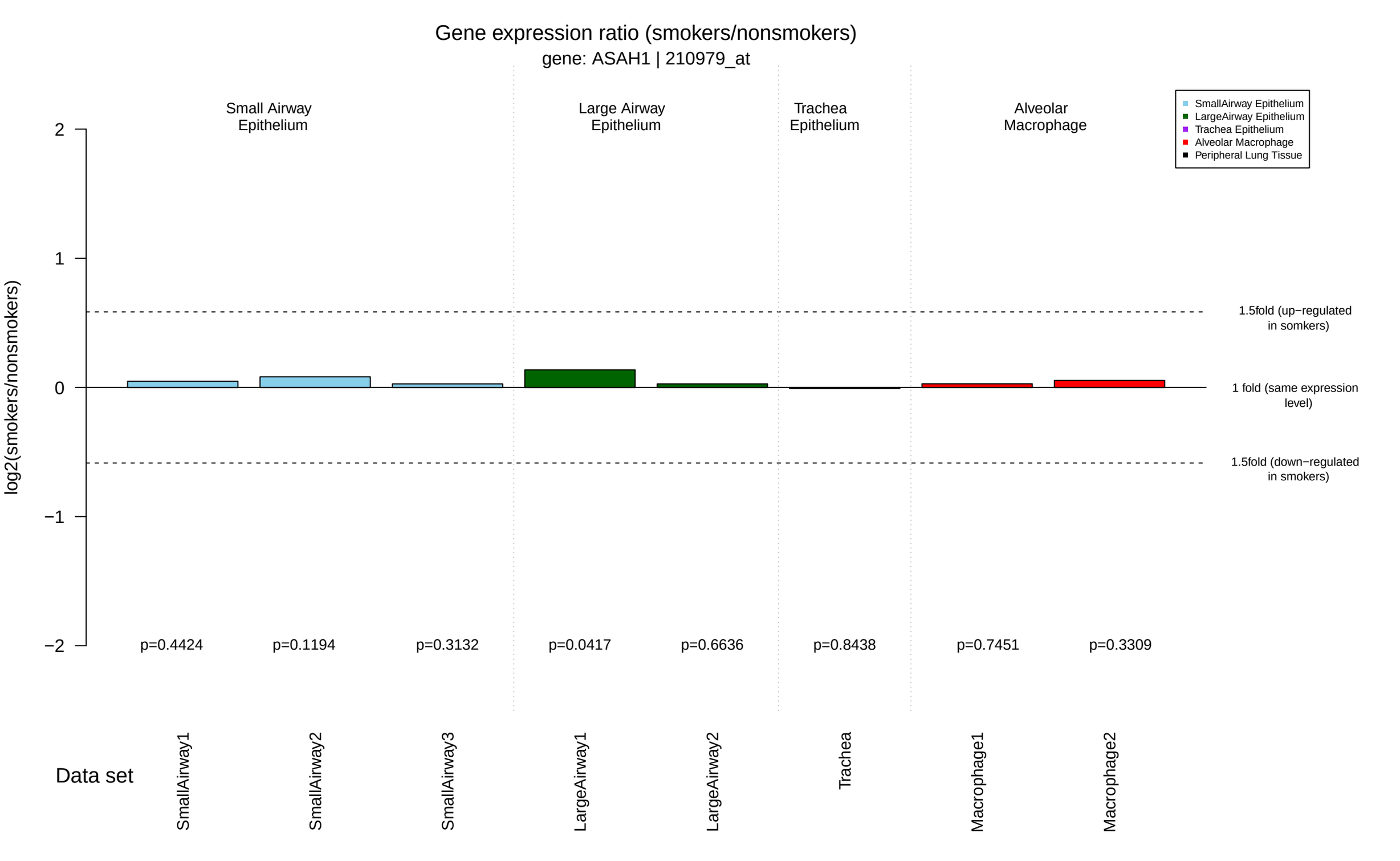
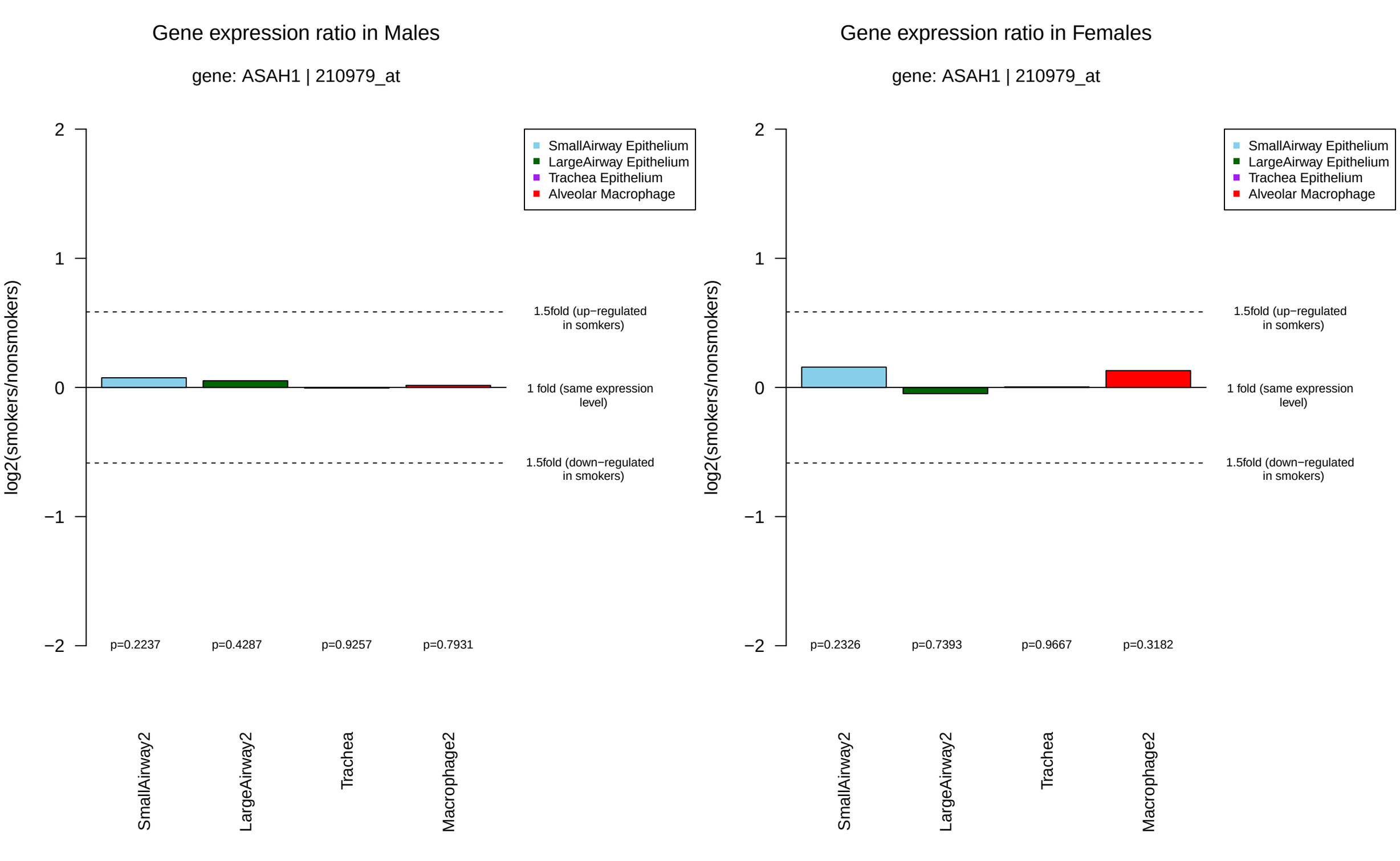
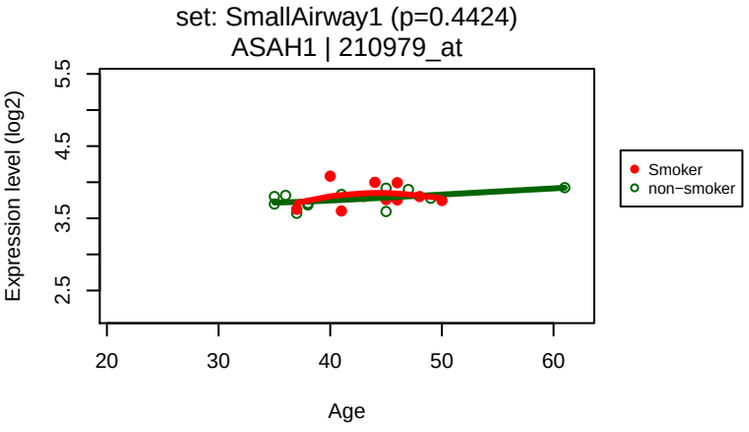
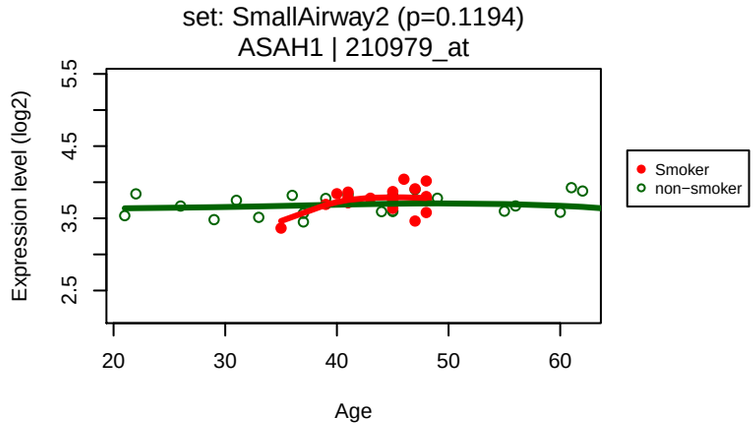
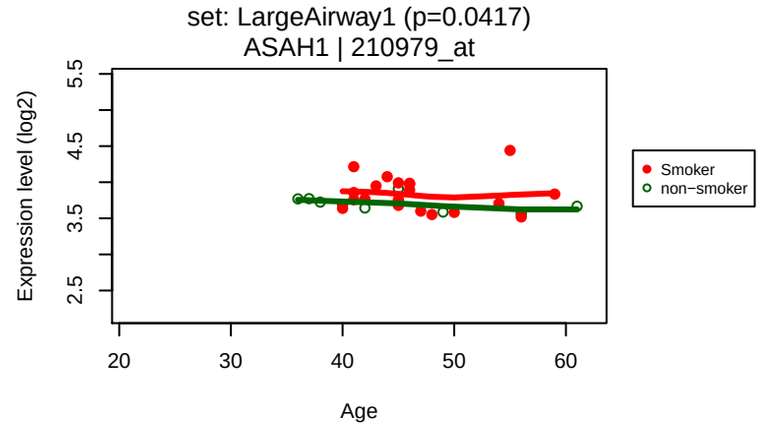

Gene expression of smokers and nonsmokers
gene: ASAH1 | 210979_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4424
p=0.1194
p=0.3132
p=0.0417
p=0.6636
p=0.8438
p=0.7451
p=0.3309
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

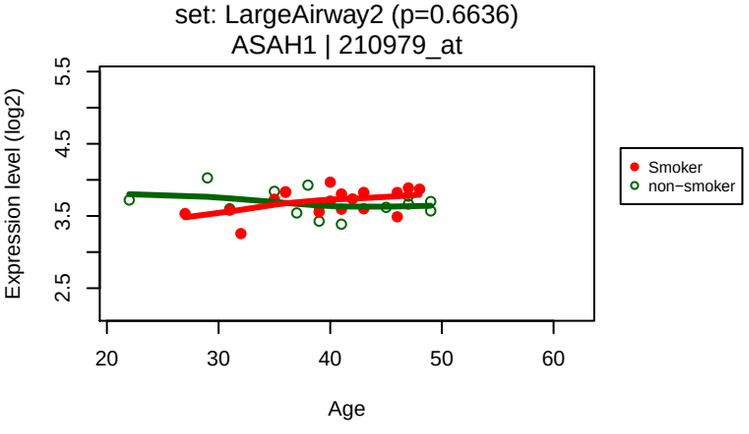

Gene expression ratio (smokers/nonsmokers)
gene: ASAH1 | 210979_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4424
p=0.1194
p=0.3132
p=0.0417
p=0.6636
p=0.8438
p=0.7451
p=0.3309
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

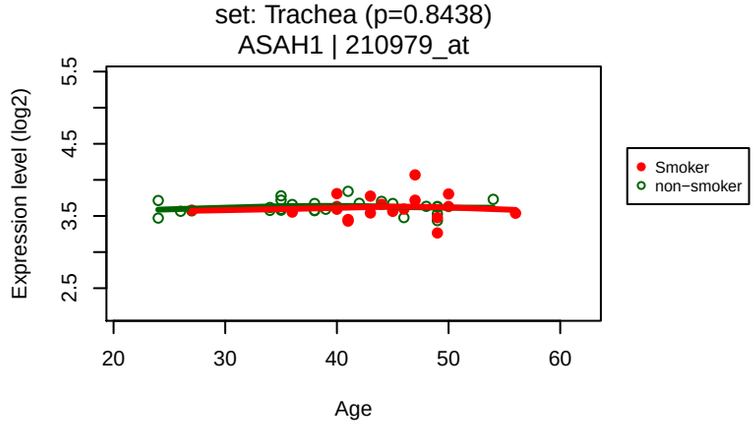

Gene expression ratio in Males
gene: ASAH1 | 210979_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2237
p=0.4287
p=0.9257
p=0.7931
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ASAH1 | 210979_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2326
p=0.7393
p=0.9667
p=0.3182
SmallAirway2
LargeAirway2
Trachea
Macrophage2

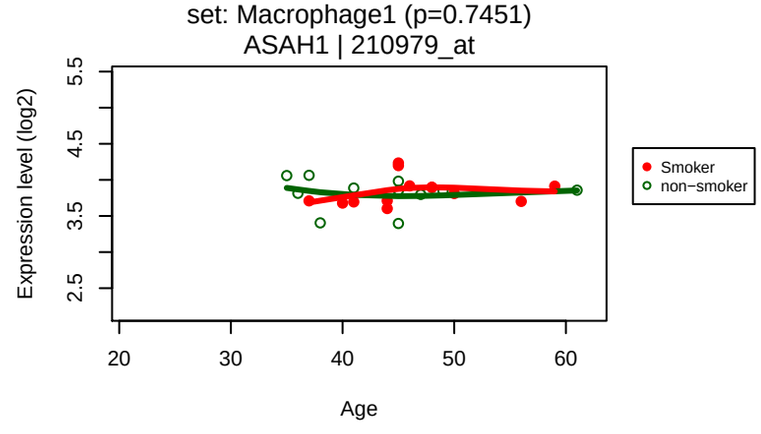

set: SmallAirway1 (p=0.4424)
ASAH1 | 210979_at
Expression level (log2)
Age
Smoker
non−smoker

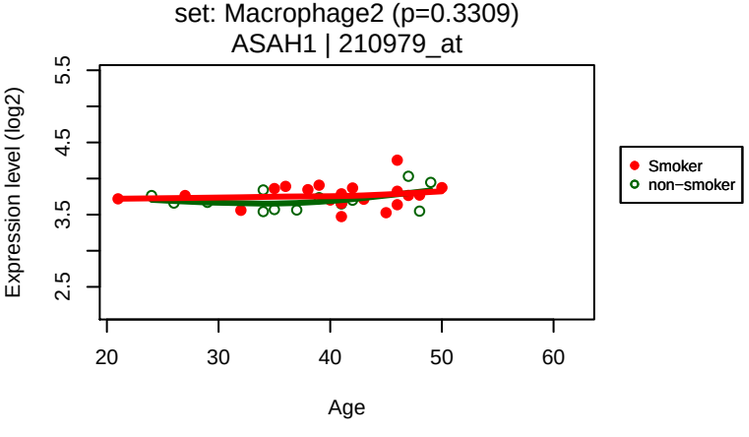

set: SmallAirway2 (p=0.1194)
ASAH1 | 210979_at
Expression level (log2)
Age
Smoker
non−smoker

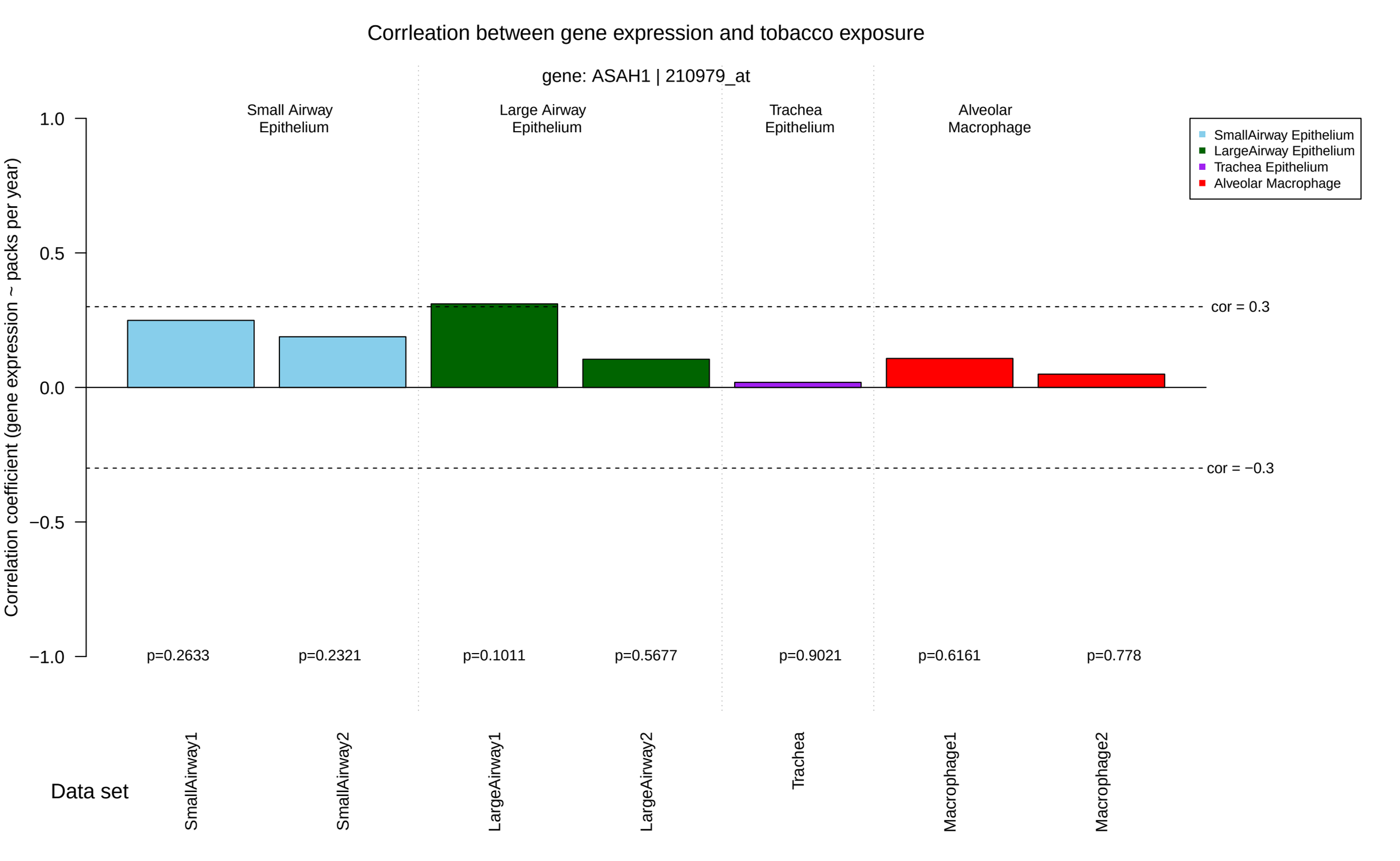

set: LargeAirway1 (p=0.0417)
ASAH1 | 210979_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6636)
ASAH1 | 210979_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8438)
ASAH1 | 210979_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7451)
ASAH1 | 210979_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3309)
ASAH1 | 210979_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ASAH1 | 210979_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2633
p=0.2321
p=0.1011
p=0.5677
p=0.9021
p=0.6161
p=0.778
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2